



Chapter:- SOLUTION



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- Q1-** Phenol associates in benzene to form a dimer $(C_6H_5OH)_2$. The freezing point of a solution containing 5 g of phenol in 250 g of benzene is lowered by $0.70^\circ C$. Calculate the degree of association of phenol in benzene. (K for benzene = 5.12 K m^{-1})

- Q2-** 1.5 g of $Ba(NO_3)_2$ dissolved in 100 g of water shows a depression in freezing point equal to $0.28^\circ C$. What is the percentage dissociation of the salt? (K for water = 1.86 K / m and molar mass of $Ba(NO_3)_2 = 261$)

- Q3-** Calculate the difference between the boiling points of 0.2 m Na_2SO_4 and 0.5 m glucose assuming complete dissociation of Na_2SO_4 (K_b for water = 0.52 K m^{-1})



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- Q4- The freezing point of a solution containing acetic acid in 20.0 g of benzene is lowered by 0.45°C . Calculate the degree of association of acetic acid in benzene. K_f benzene = $5.12 \text{ K mol}^{-1} \text{ kg}$

- Q5- Calculate the normal freezing point of a sample of sea water containing 3.8% NaCl and 0.12% MgCl_2 by mass. (K_f for water = 1.86 K m^{-1}).

- Q6- Calculate the molality of NaCl solution whose elevation in boiling point is equal to the depression in freezing point of 0.25 m sodium carbonate solution in water assuming complete dissociation of salts. ($K_f = 1.86 \text{ K m}^{-1}$, $K_b = 0.52 \text{ K m}^{-1}$).



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Q7- 0.01m aqueous solution of $K_3[Fe(CN)_6]$ freezes at $-0.062^\circ C$. What is the apparent percentage of dissociation? ($K_f = 1.86 K\ kg\ mol^{-1}$)

Q8- Calculate the amount of KCl which must be added to 1 kg of water so that the freezing point is depressed by 2K. (K for water = $1.86 K\ kg\ mol^{-1}$)

Q9- The freezing point of a solution containing 5g of benzoic acid ($M = 122\ gmol^{-1}$) in 35g of benzene is depressed by 2.94 K. What is the percentage association of benzoic acid if it forms a dimer in solution? (K_f for benzene = $4.9 K\ kg\ mol^{-1}$)



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Q10- 0.01 M solutions of KCl and BaCl₂ are prepared in water. The freezing point of KCl is found to be -2°C. What freezing point would you expect for BaCl₂ solution assuming both KCl and BaCl₂ to be completely ionized?

Q11- Calculate the freezing point of an aqueous solution containing 10.50 g of MgBr₂ in 200 g of water. (Molar mass of MgBr₂ = 184, K for water = 1.86 kg mol⁻¹)
